

Lab W15 Assignment: Style-Conditioned LoRA

Fine-Tuning of Qwen2.5-0.5B-Instruct

Qwen2.5-0.5B-Instruct

请用唐僧的风格回答下面问题，什么是友谊



在唐僧的风格下，友谊是一种深厚的情感和精神上的支持。它不仅仅是相互帮助和鼓励，更是彼此之间的信任和理解。在面对困难和挑战时，友谊如同一座桥梁，帮助我们找到前进的方向。它让我们在孤独和困难中找到安慰，增强彼此之间的联系。友谊是心灵的慰藉，它让我们的生活更加丰富多彩。在佛教中，友谊被描述为“菩提树”，它代表了超越自我，达到涅槃的最高境界。所以，无论在日常生活中还是在精神上，唐僧都强调友谊的重要性。



Qwen2.5-0.5B-Instruct + LoRA

请用唐僧的风格回答下面问题，什么是友谊



阿弥陀佛，若能多真诚相处，多体谅对方，便会更稳妥。



请用孙悟空的风格回答下面问题，什么是友谊



俺老孙觉得，真心付出，多分担责任，咱就能把事办明白。



请用猪八戒的风格回答下面问题，什么是友谊



老猪我觉得，真心支持、分担困难，就差不多啦。



1. Overview

In this lab, you will use **LLaMA-Factory** to perform **LoRA** fine-tuning on **Qwen2.5-0.5B-Instruct**, so that the model can respond in **multiple language styles** of your own choosing (e.g., Classical Chinese, Shakespearean English, Lu Xun style, anime character tone, Northeastern dialect, etc.).

By the end of this lab, you should be able to:

- Understand how LoRA adapts a pre-trained LLM with a small number of trainable parameters.
- Construct a custom instruction-tuning dataset from scratch.
- Run an end-to-end fine-tuning pipeline using LLaMA-Factory.
- Critically evaluate the behavioral difference between the base model and your fine-tuned model.

2. Background

- **Base Model:** [Qwen2.5-0.5B-Instruct](#) — a small, instruction-tuned LLM suitable for quick

experimentation.

- **Method:** [LoRA \(Low-Rank Adaptation\)](#) — freezes the original model weights and injects trainable low-rank matrices into selected layers.
- **Framework:** [LLaMA-Factory](#) — a unified framework for fine-tuning LLMs with rich support for LoRA, QLoRA, full-parameter tuning, etc.

3. Task Description

3.1 Choose Your Styles

Select **2–3 distinct language styles** that you would like your model to speak in. Examples (not exhaustive):

- Classical Chinese / Literary Chinese (文言文)
- Shakespearean English
- A specific author's voice (e.g., Lu Xun, Hemingway)
- A fictional character's tone (e.g., a tsundere anime character, a pirate)
- Regional dialect (e.g., Cantonese-style, Northeastern Mandarin)
- Domain-specific tone (e.g., legal contract, academic paper, news anchor)

Your styles should be **clearly distinguishable** from one another so that the effect of fine-tuning is observable.

3.2 Construct the Dataset

You are required to build your training data **by modifying and extending the `identity` dataset** that ships with LLaMA-Factory (`data/identity.json`).

Requirements:

- Keep the **conversational format** compatible with LLaMA-Factory (`alpaca` or `sharegpt`).
- For each chosen style, create a sufficient number of (`instruction`, `output`) pairs that demonstrate that style.
- Make sure the dataset **explicitly signals the target style** in the instruction (e.g., system prompt or a style tag), so the model learns *conditional* style generation rather than collapsing to a single voice.
- Document how many samples you created per style and how you generated them (handwritten, distilled from a larger LLM, scraped, etc.).

3.3 Fine-Tune with LoRA

Using LLaMA-Factory, run LoRA fine-tuning on your dataset. You should:

- Configure the LoRA hyperparameters (`rank`, `alpha`, `target_modules`, `dropout`).
- Choose appropriate training hyperparameters (`learning_rate`, `batch_size`, `epochs`, `warmup`, etc.).
- Log the **training loss curve** for later analysis.

3.4 Inference and Comparison

After training:

- Load the LoRA adapter and run inference on a set of test prompts.
- Run the **same prompts** on the **base Qwen2.5-0.5B-Instruct** (without LoRA) for comparison.
- Identify both **successful cases** and **failure cases**.

4. Deliverables

4.1 Lab Report (PDF)

Your report must include **at least** the following three sections:

Section	Required Content
Data Construction & Processing	Chosen styles, modifications to the <code>identity</code> dataset, number of samples per style, sample examples, and rationale for your data design.
Training Hyperparameters & Curve Analysis	Full list of hyperparameters used; screenshot of the loss curve; analysis of convergence, overfitting, and any tuning decisions you made.
Effect Comparison & Case Study	Side-by-side comparison of base model vs. fine-tuned model outputs on the same prompts; analysis of typical successes and failures.

A good report does **not** simply paste screenshots — it analyzes *why* certain things worked or did not.

4.2 Demo Images

Submit at least the following two types of images:

1. **Conversation Screenshots:** Real interactions with your fine-tuned model demonstrating each chosen style.
2. **Before/After Comparison Image:** For the same prompt, show the base model's output next to the LoRA-tuned model's output.

5. Tips

- Start with a **small dataset** (e.g., 50–100 samples per style) to validate the pipeline end-to-end before scaling up.
- Use a **system prompt or style tag** in your instructions so the model can distinguish between styles at inference time.
- Be careful about **overfitting** — with a 0.5B model and a small dataset, the model may simply memorize. Watch the loss curve and try varying `epochs` and `rank`.
- Keep a **clean folder structure**: separate `data/`, `configs/`, `outputs/`, and `report/`.

- A usable environment configuration

```
git clone --depth 1 https://github.com/hiyouga/LlamaFactory.git
cd LlamaFactory
conda create -n llama_factory python=3.11
conda activate llama_factory
pip install torch==2.6.0 torchvision==0.21.0 torchaudio==2.6.0 --index-url
https://download.pytorch.org/whl/cu124 #gpu 版本
pip install -e .
pip install -r requirements/metrics.txt
pip install addic -U
llamafactory-cli webui
```

6. Resources

- LLaMA-Factory: <https://github.com/hiyouga/LLaMA-Factory>.
- Qwen2.5-0.5B-Instruct: <https://huggingface.co/Qwen/Qwen2.5-0.5B-Instruct>
- LoRA paper: <https://arxiv.org/abs/2106.09685>
- LLaMA-Factory data format docs: <https://github.com/hiyouga/LLaMA-Factory/blob/main/data/README.md>

7. Submission

- **Format:** Submit a single `.zip` archive named `lab_lora_<your_name>_<student_id>.zip` containing:
 - `report.pdf`
 - `demo/` (folder of demo images)
 - `data/` (your constructed dataset, in LLaMA-Factory format)

Good luck, and have fun teaching a tiny LLM to speak in many voices!